Assignment on RNN

Assignment 1

1. Create a simple RNN layer using **TensorFlow/Keras**.
2. Pass a dummy input (random tensor) through the RNN and print the **output** and **hidden state**.

Assignment 2

Train an RNN to predict the next number in a simple sequence. The model should learn to **predict the next number** in the sequence. For input [7, 8, 9], output should be close to 10.

Assignment 3

Define an **RNN model** with:

- An **embedding layer**
- An **RNN layer** (hidden size = 16)
- A **fully connected layer** for output.

Train it on the "hello world" dataset to predict the next character. Plot the **loss curve**.

Assignment 4

To implement an RNN-based model that predicts future stock prices based on historical data.

Dataset : https://www.kaggle.com/datasets/tarunpaparaju/apple-aapl-historical-stock-data

Download historical stock price data (e.g., Apple, Google, or TCS) from the above link.

Choose the **"Close"** price column as the target variable.

Normalize the data (e.g., MinMax scaling to [0,1]) to improve RNN training.

Create input-output pairs:

- Example (window = 60 days):
  - Input: prices from day 1 → day 60;
  - Output: price of day 61

Define an RNN model with:

- Input layer

- RNN layer with hidden units
- Fully connected output layer (predict next price)

Use **MSE loss function** and **Adam optimizer**. Train the model for 50–100 epochs. Plot the **training loss curve**. Predict stock prices on the **test set**. Plot actual vs predicted prices.